

Forensic Microscopy

Lab 5: Uniaxial Interference figures

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Where can you find the slides?

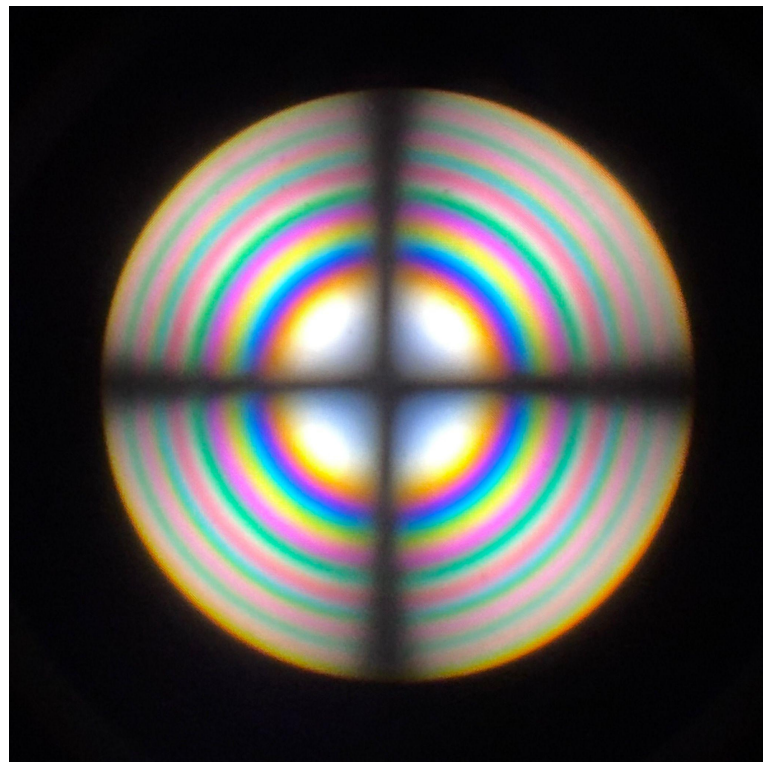


- <https://aritra-evolves.github.io/FIVS215.html>
- Canvas

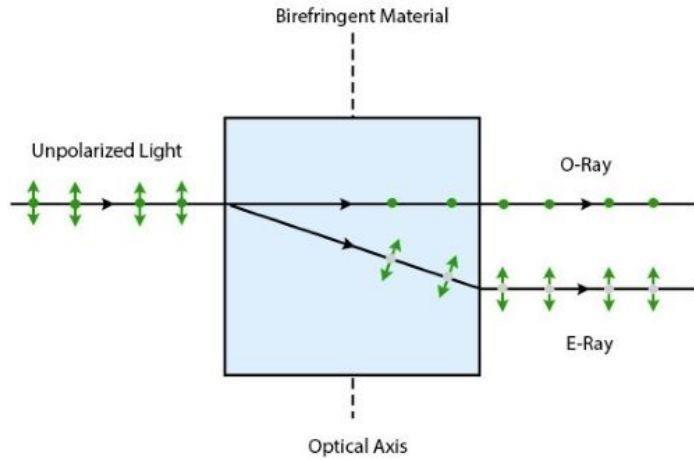
(Grab a Microscope from the black closet, set up kohler)

Today's Agenda

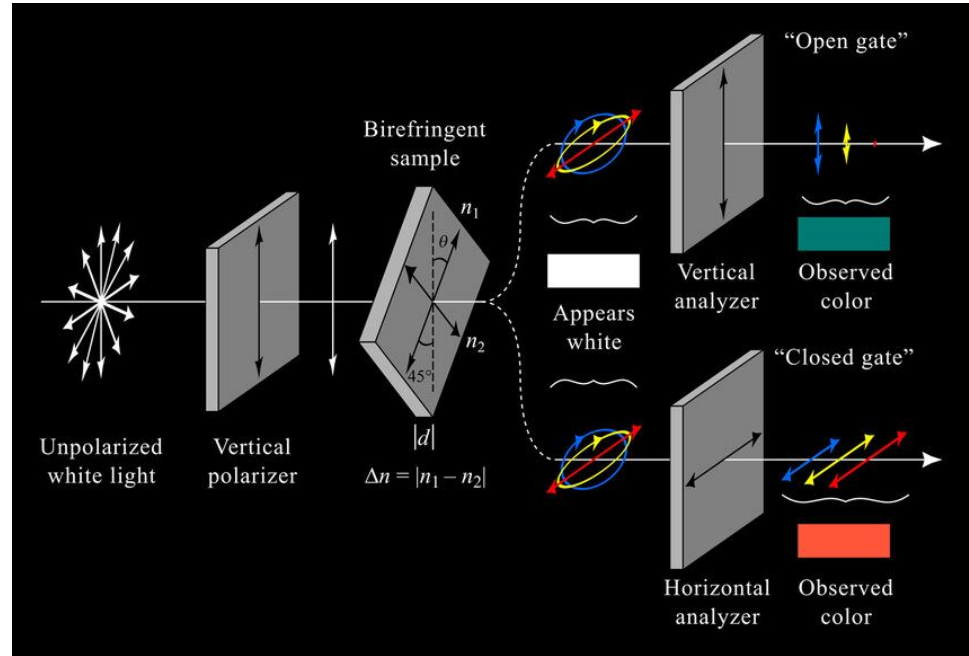
- ❖ Basic concepts
- ❖ Interference Figures
- ❖ Viewing
- ❖ Practice



Birefringence



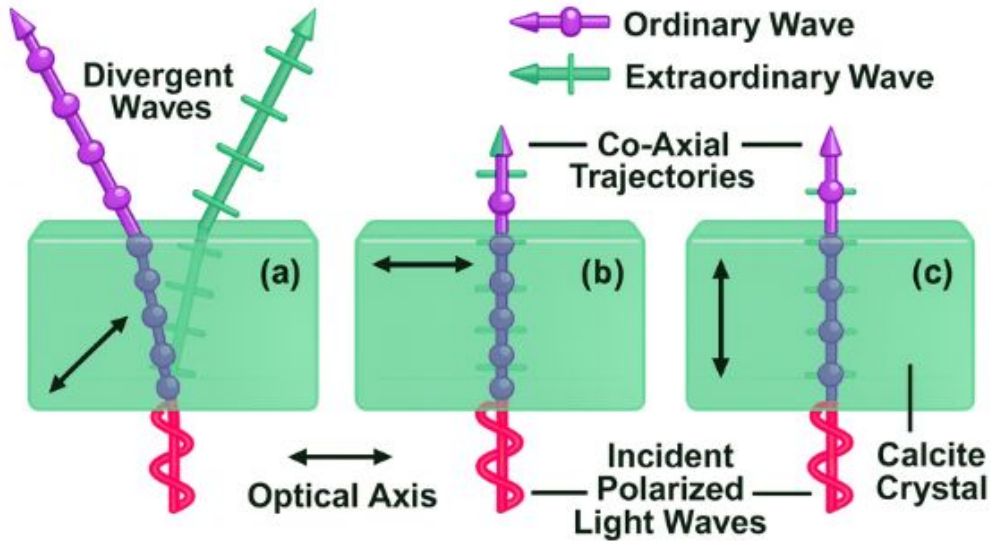
<https://www.gophotonics.com/community/what-is-birefringence>



https://www.researchgate.net/figure/Color-online-A-schematic-representation-of-the-principle-of-polarization-filtered_fig1_362400155

Optical Axis

Separation of Light Waves by a Birefringent Crystal



- The optic axis of a crystal is the direction in which the light propagates through the crystal without facing **double refraction**.
- All the light waves that are parallel to this axis do not undergo double refraction. In other words, the light beam passes in this direction with a constant speed that does not depend on the polarization.

Positive vs. Negative:

$$\epsilon - \omega = \text{optic sign}$$

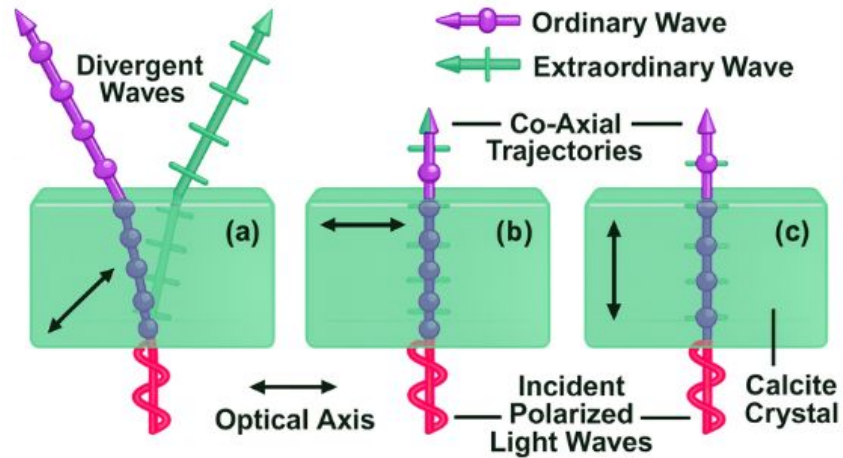
ϵ (epsilon): Refractive index of the Extraordinary ray (n_e)

ω (omega): Refractive index of the Ordinary ray (n_o)

Positive: $n_o < n_e$

Negative: $n_e < n_o$

Separation of Light Waves by a Birefringent Crystal



Interference Figures

- Allow us to identify minerals
 - Specifically those with tetragonal and hexagonal crystalline structures
- Based on the vibration direction of the slow component of the wave plate
- Tells us the speed at which epsilon is vibrating

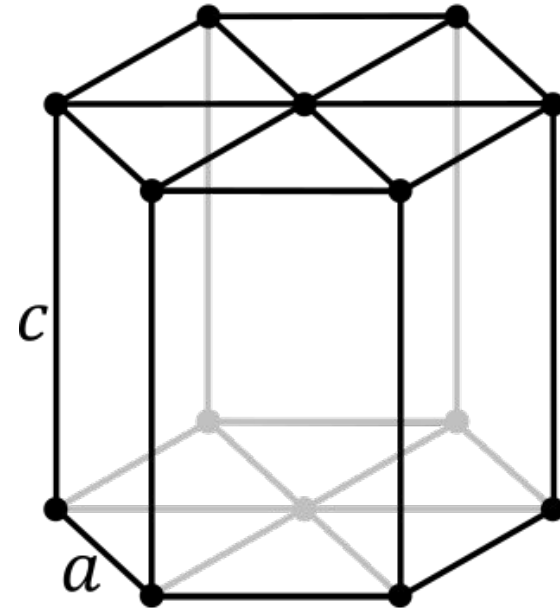
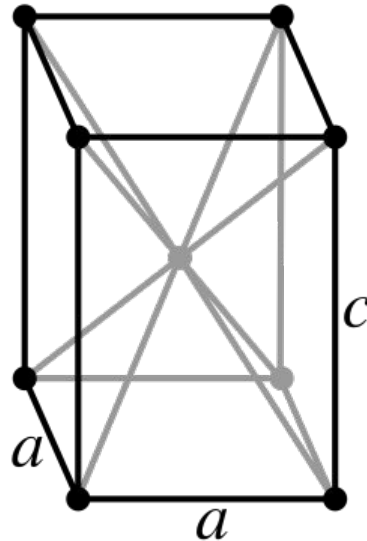


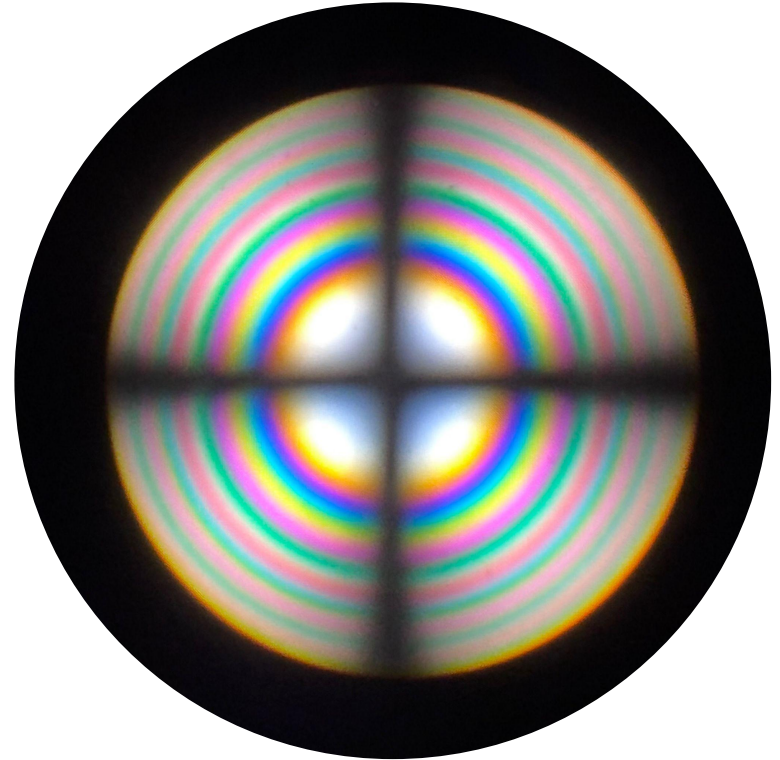
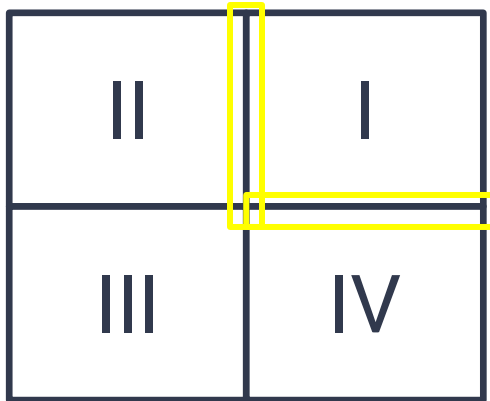
Image: https://commons.wikimedia.org/wiki/File:Hexagonal_lattice.svg

Image: https://en.wikipedia.org/wiki/Tetragonal_crystal_system

Centering

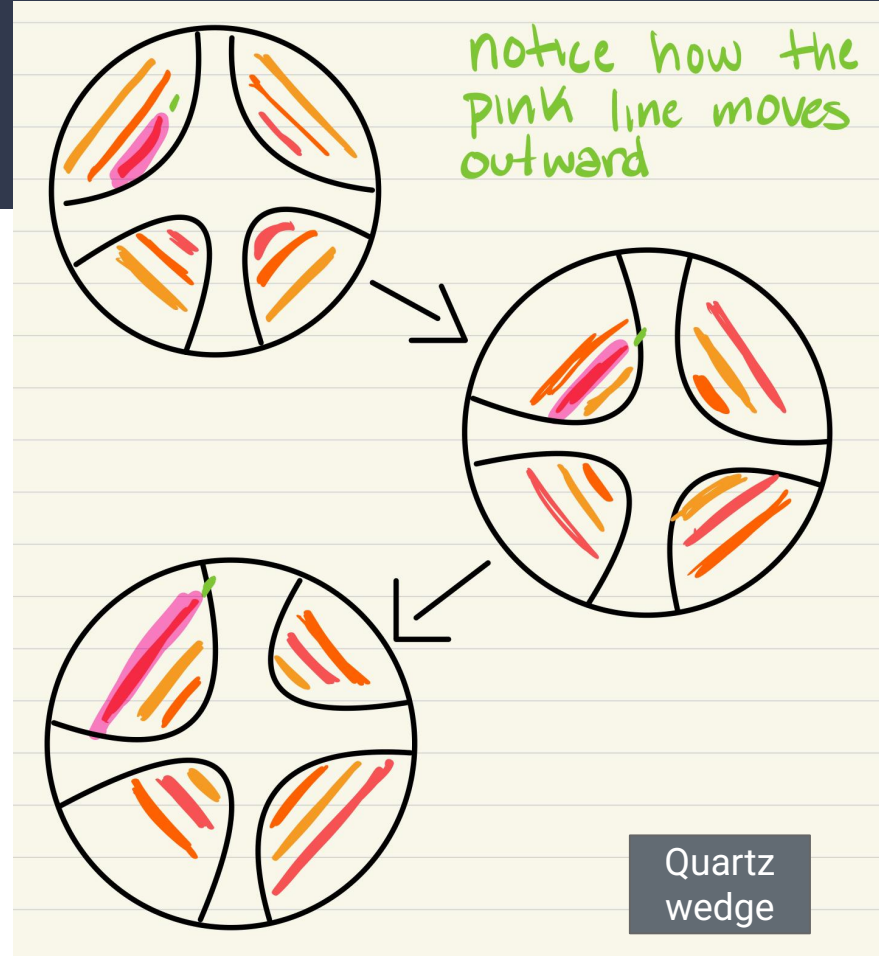
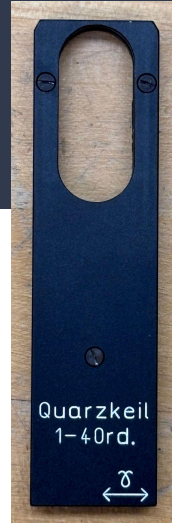
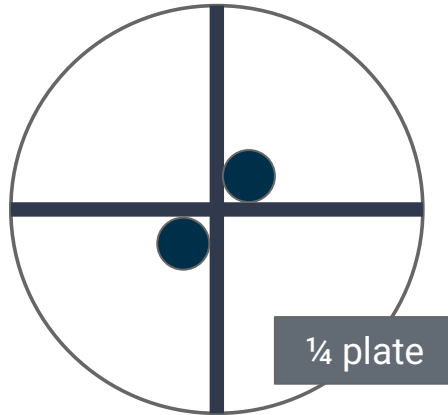
- As centered as possible is IDEAL
 - Not always possible
- Use movement of the isogyres to tell you which quadrant is which
- As you rotate the stage you can tell which quadrant is which by where each isogyre is

[video](#)



Viewing

- Viewing using **conoscopy**
 - Utilizes the Amici-Bertrand Lens
 - Shows the Back Focal Plane
- Looking down the **optic axis**
 - Plane that is parallel to the light source
- Rotating the stage helps us view this
 - **Spindle** stages or **goniometer** stages
- Three ways
 - Quarter Wave Plate
 - Full Wave Plate
 - Quartz Wedge

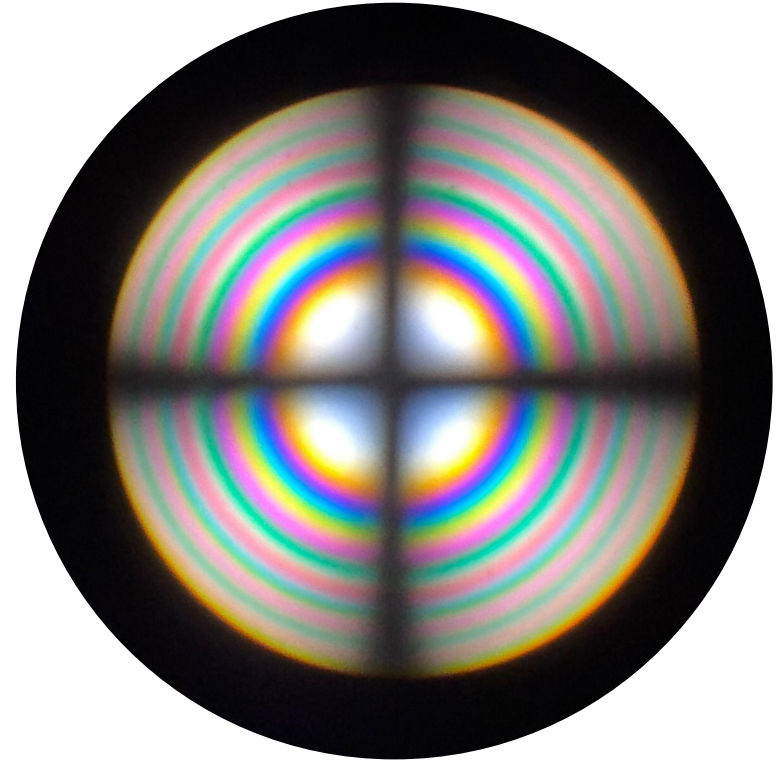
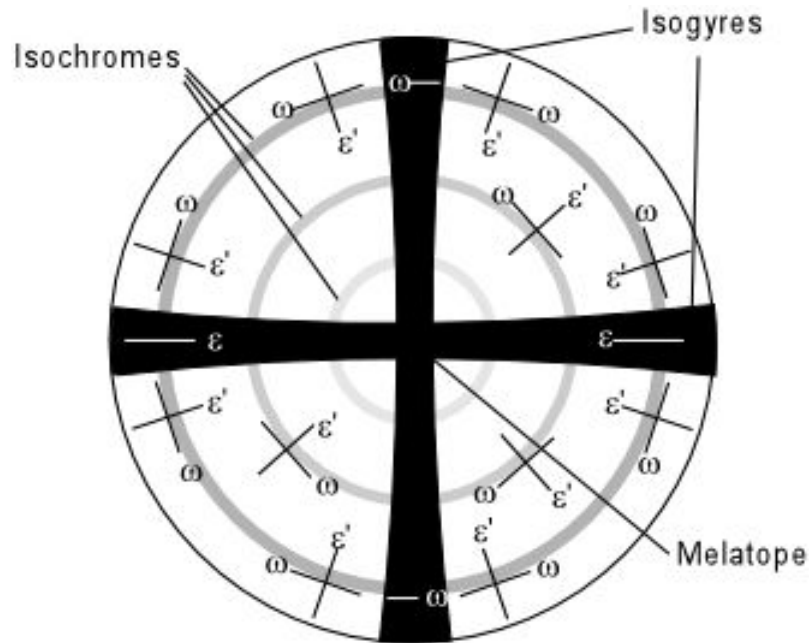


Figures: Alli Findley

Practice

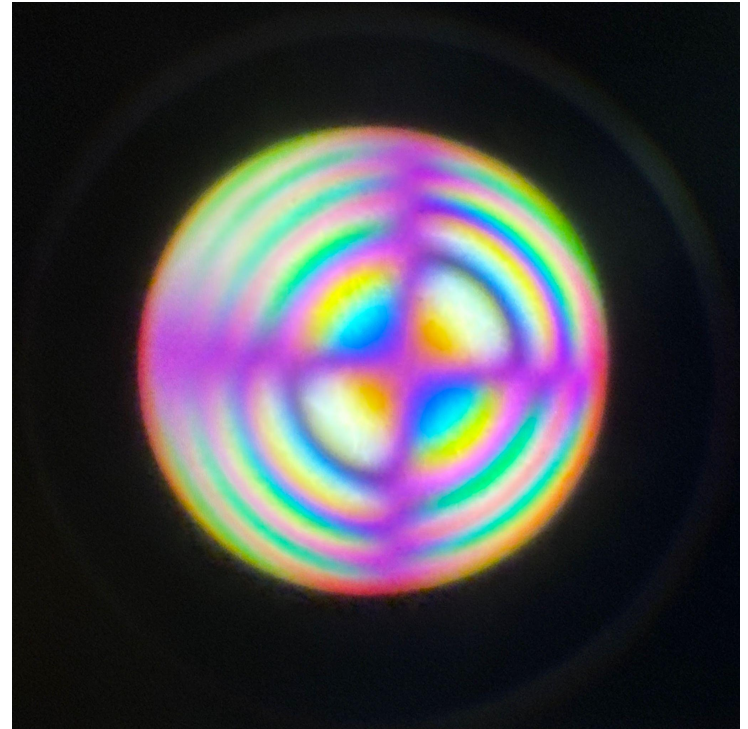
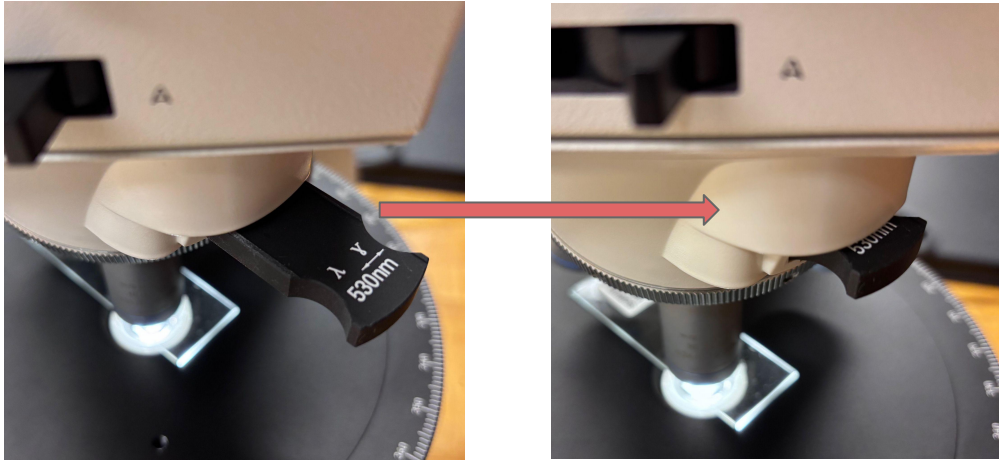
1. Get a Sodium Nitrate Sample from the Back
2. Pull Out your Wave Plate
3. Set up Kohler on 10x
4. Cross your Polars
5. Look for Areas that Remain Gray when Rotating
6. Swing in a Higher Objective
7. Open your Substage Aperture
8. Swing in the Bertrand Lens
9. Observe the Roman Cross
10. Insert a Compensator to Observe the Optic Sign

Steps 4-9 – What should you see?



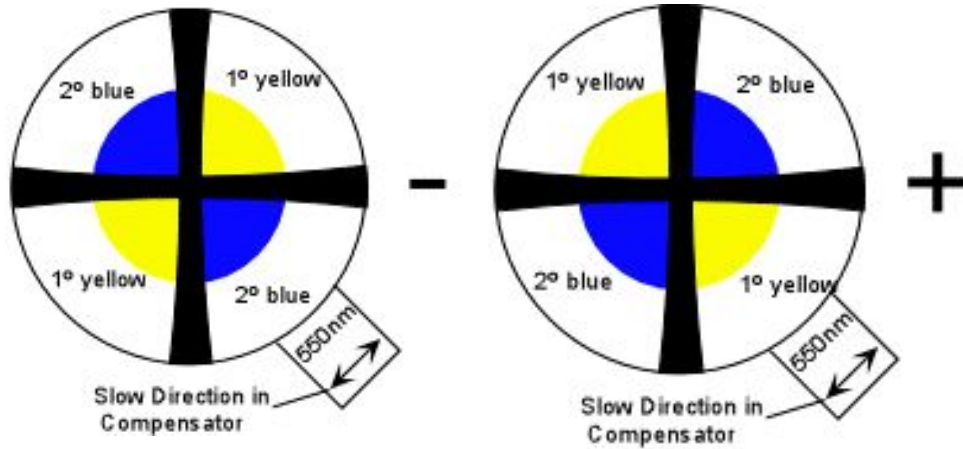
Add in a Compensator

- Add in the Compensator
- You should see lots of colors
- Focus on the colors closest to the center of the cross (melatope)
 - Should be blue and yellow/orange



How to determine Positive vs. Negative ?

BURP: blue *upper right* positive



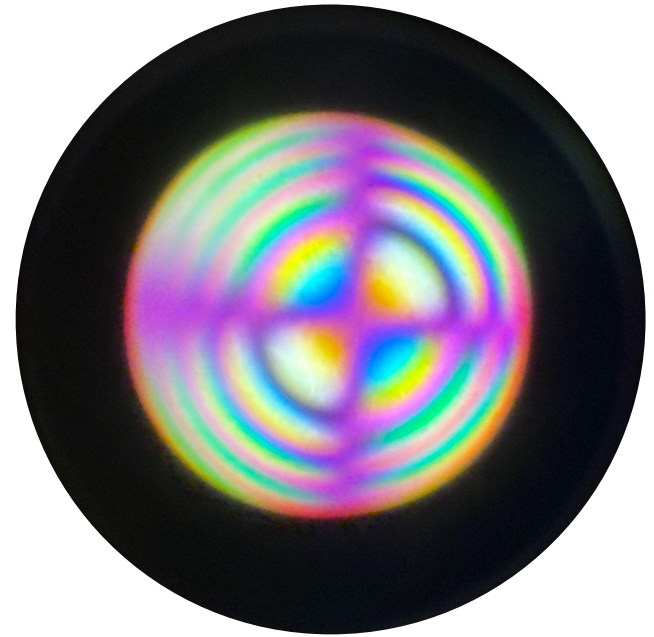
$\epsilon - \omega = \text{optic sign}$

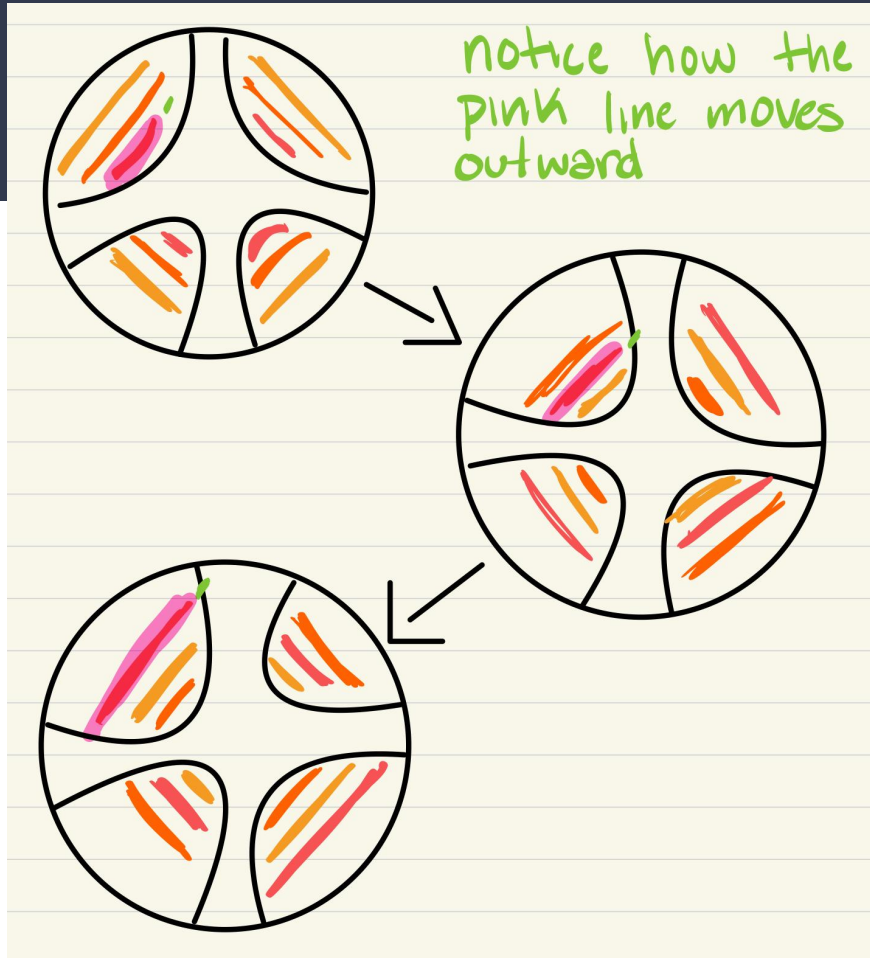
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When you insert the quartzkeil slowly:

Isochromes move **toward** the melatopes in NE/SW

Positive (+)

Isochromes move **away** from the melatopes in NE/SW

Negative (-)

See you next week!